GENG5507 Probability and Statistics Cheatsheet

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Use this after learning the concepts in the study map. The worked examples show how the formulas are applied.

1. Translate the wording

Words	Symbol
not A	A^c
A or B, including both	$A \cup B$
A and B	$A \cap B$
A, given B	$P(A \mid B)$
fails by t	$F(t) = P(T \leq t)$
survives beyond t	$R(t) = P(T > t)$
between a and b	$F(b) - F(a)$
at least one	$1 - P(\text{none})$
probability from a boundary	CDF: R prefix p
boundary from a probability	quantile: R prefix q

2. Event rules

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A \mid B)P(B)$$

For independent events,

$$P(A \cap B) = P(A)P(B).$$

For independent $A_1, ..., A_n$,

$$P(\text{at least one}) = 1 - \prod_{i=1}^{n} P(A_i^c).$$

If each has probability p,

$$P(\text{at least one}) = 1 - (1-p)^n.$$

Law of total probability: split B into the cases where A happens and where it does not. For $0 < P(A) < 1$,

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

Weight each conditional probability by the probability of its case; $P(B \mid A) + P(B \mid A^c)$ alone is not the rule. Equivalently, $P(B) = P(B \cap A) + P(B \cap A^c)$.

More generally, if $B_1, ..., B_k$ form a partition (mutually exclusive, exhaustive cases with positive probabilities),

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i).$$

Bayes:

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_i P(A \mid B_i)P(B_i)}.$$

Do not confuse:

- disjoint: $P(A \cap B) = 0$;
- independent: $P(A \cap B) = P(A)P(B)$.

Positive-probability disjoint events are not independent.

3. Random-variable functions

Object	Discrete	Continuous
Probability model	PMF $p(x) = P(X = x)$	PDF $f(x)$, a density
Total probability	$\sum_x p(x) = 1$	$\int f(x)\,\mathrm{dx} = 1$
Interval probability	Sum relevant masses	$\int_a^b f(x)\,\mathrm{dx}$
CDF	$F(x) = P(X \le x)$	$F(x) = \int_{-\infty}^{x} f(u)\,\mathrm{du}$
Mean	$\sum_x xp(x)$	$\int xf(x)\,\mathrm{dx}$

$$\mathrm{Var}(X) = E[X^2] - E[X]^2, \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

For a continuous variable,

$$P(a < X < b) = F(b) - F(a), \qquad P(X = x) = 0.$$

4. Reliability and hazard

Let $T \geq 0$ be the time to failure. The **failure function** (CDF) is

$$F(t) = P(T \leq t),$$

the probability that the item **has failed by time t**. The **reliability function** (survival function) is

$$R(t) = P(T > t),$$

the probability that the item **has not failed by time t** and survives beyond it. These are complementary events, so

$$F(t) + R(t) = 1, \qquad R(t) = 1 - F(t).$$

For a continuous lifetime, the other functions are:

Function	What it is	Meaning
$f(t) = F'(t) = -R'(t)$	Failure-time density (PDF)	Describes how concentrated failures are near t. For small Δt, $f(t)\Delta t \approx P(t < T \leq t + \Delta t)$.
$h(t) = \frac{f(t)}{R(t)}$	Hazard rate	Instantaneous failure rate at age t **among items that have survived to t.**
$H(t) = \int_0^t h(u)\,\mathrm{du}$	Cumulative hazard	Total hazard accumulated from time 0 to t.

Thus, for small Δt,

$$P(t < T \leq t + \Delta t \mid T > t) \approx h(t)\Delta t.$$

The functions describe the same lifetime distribution and can be converted using

$$R(t) = e^{-H(t)}, \qquad f(t) = h(t)R(t).$$

The mean time to failure is the expected lifetime, or equivalently the area under the reliability curve:

$$\mathrm{MTTF} = E[T] = \int_0^\infty t f(t)\,\mathrm{dt} = \int_0^\infty R(t)\,\mathrm{dt}.$$

Both $f(t)$ and $h(t)$ are densities or rates, not probabilities. If time is measured in hours, their units are $\frac{1}{\mathrm{hour}}$.

Multicomponent systems

At time t, let component i have reliability $R_{i(t)}$. If component failures are independent:

System	It works when	System reliability
Series	Every component works	$R_{\text{series}}(t) = \prod_{i=1}^{m} R_{i(t)}$
Parallel	At least one component works	$R_{\text{parallel}}(t) = 1 - \prod_{i=1}^{m}\left[1 - R_{i(t)}\right]$

For m independent components with identical reliability $R(t)$,

$$R_{\text{series}}(t) = R(t)^m, \qquad R_{\text{parallel}}(t) = 1 - [1 - R(t)]^m.$$

For either system, $F_{\text{sys}}(t) = 1 - R_{\text{sys}}(t)$. The product formulas do not apply directly when component failures are dependent.

For an independent series system, cumulative hazards and hazard rates add:

$$H_{\text{series}}(t) = \sum_{i=1}^{m} H_{i(t)}, \qquad h_{\text{series}}(t) = \sum_{i=1}^{m} h_{i(t)}.$$

If component i has a constant exponential failure rate λ_i, then

$$\lambda_{\text{series}} = \sum_{i=1}^{m} \lambda_i, \qquad R_{\text{series}}(t) = \exp\left(-t \sum_{i=1}^{m} \lambda_i\right).$$

5. Choose a distribution

Model	Use when	Mean	Standard deviation
Binomial (n, p)	Success count in fixed n independent trials	np	$\sqrt{np(1-p)}$
Poisson $(\mu = \lambda t)$	Event count in an interval	μ	$\sqrt{\mu}$
Uniform $[a, b]$	All positions in a bounded interval equally likely	$\frac{a+b}{2}$	$\frac{b-a}{\sqrt{12}}$
Normal (μ, σ^2)	Symmetric measurements	μ	σ
Lognormal (μ, σ^2)	$\ln T \sim N(\mu, \sigma^2)$	$e^{\mu+\frac{\sigma^2}{2}}$	$e^{\mu+\frac{\sigma^2}{2}}\sqrt{e^{\sigma^2}-1}$
Exponential (λ)	Constant-hazard lifetime	$\frac{1}{\lambda}$	$\frac{1}{\lambda}$
Weibull (β, η)	Flexible monotone-hazard lifetime	$\eta\Gamma\left(1+\frac{1}{\beta}\right)$	$\eta\sqrt{\Gamma\left(1+\frac{2}{\beta}\right)-\Gamma\left(1+\frac{1}{\beta}\right)^2}$
Gamma, shape k, rate λ	Positive staged waiting time	$\frac{k}{\lambda}$	$\frac{\sqrt{k}}{\lambda}$

CDF, reliability, hazard and MTTF at a glance

For continuous lifetimes, $R(t) = 1 - F(t)$ and $h(t) = \frac{f(t)}{R(t)}$ wherever $R(t) > 0$. All scale, shape and rate parameters below are positive. Write $\phi(z) = \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}}$ and $\Phi(z) = P(Z \le z)$ for the standard normal PDF and CDF.

Distribution	CDF $F(t)$	Reliability $R(t)$	Hazard $h(t)$	MTTF $E[T]$
Uniform $[a, b]$, $0 \le a < b$; $a \le t < b$	$\frac{t-a}{b-a}$	$\frac{b-t}{b-a}$	$\frac{1}{b-t}$	$\frac{a+b}{2}$
Normal (μ, σ^2); $z = \frac{t-\mu}{\sigma}$	$\Phi(z)$	$1 - \Phi(z)$	$\frac{\phi(z)}{[\sigma(1-\Phi(z))]}$	μ (mean; lifetime approximation only)
Lognormal (μ, σ^2); $z = \frac{\ln t-\mu}{\sigma}$, $t > 0$	$\Phi(z)$	$1 - \Phi(z)$	$\frac{\phi(z)}{[t\sigma(1-\Phi(z))]}$	$e^{\mu+\frac{\sigma^2}{2}}$
Exponential (λ); $t \ge 0$	$1 - e^{-\lambda t}$	$e^{-\lambda t}$	λ	$\frac{1}{\lambda}$
Weibull (β, η); $t > 0$	$1 - e^{-\left(\frac{t}{\eta}\right)^\beta}$	$e^{-\left(\frac{t}{\eta}\right)^\beta}$	$\left(\frac{\beta}{\eta}\right)\left(\frac{t}{\eta}\right)^{\beta-1}$	$\eta\Gamma\left(1+\frac{1}{\beta}\right)$

Distribution	CDF $F(t)$	Reliability $R(t)$	Hazard $h(t)$	MTTF $E[T]$
Gamma, shape k, rate λ; $t > 0$	$G_{k(\lambda t)}$	$1 - G_{k(\lambda t)}$	$\lambda^k t^{k-1} \frac{e^{-\lambda t}}{\left[\Gamma(k)\left(1-G_{k(\lambda t)}\right)\right]}$	$\frac{k}{\lambda}$

Here the gamma CDF uses

$$G_{k(y)} = \frac{1}{\Gamma(k)} \int_0^y u^{k-1} e^{-u} \, \mathrm{du}.$$

For integer $k \geq 1$, gamma reliability also has the finite-sum form

$$R(t) = e^{-\lambda t} \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!}.$$

For uniform lifetimes, $F = 0$, $R = 1$, $h = 0$ when $t < a$; $F = 1$, $R = 0$ when $t \geq b$, where hazard is undefined. For lognormal, exponential, Weibull and gamma lifetimes, $F(t) = 0$ and $R(t) = 1$ for $t \leq 0$; hazard at 0 is interpreted by its right-hand limit. The untruncated normal allows negative values, so its mean μ is only an approximate MTTF when the negative tail is negligible. The nonnegative-lifetime identity $E[T] = \int_0^\infty R(t) \, \mathrm{dt}$ does not hold for that untruncated model.

Count distributions: binomial and Poisson CDFs are sums of masses. For $x \geq 0$,

$$F_{\mathrm{bin}}(x) = \sum_{j=0}^{\min(n, \lfloor x \rfloor)} \binom{n}{j} p^{j(1-p)^{n-j}}, \qquad F_{\mathrm{Pois}}(x) = \sum_{j=0}^{\lfloor x \rfloor} \frac{\mu^j e^{-\mu}}{j!}.$$

Both CDFs are 0 for $x < 0$, and $P(X > x) = 1 - F(x)$. Their means np and μ are counts, not MTTFs. If an integer-valued variable instead represents lifetime in discrete time steps, use the **discrete hazard** $P(T = j \mid T \geq j) = \frac{p(j)}{[1-F(j-1)]}$, not the continuous hazard formula, and $E[T] = \sum_{j=0}^{\infty} P(T > j)$ for $T \geq 0$.

Binomial

$$P(X = x) = \binom{n}{x} p^{x(1-p)^{n-x}}.$$

Check: fixed n, two outcomes, constant p, independent trials.

Poisson

$$P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \qquad \mu = \lambda t.$$

Check: count, constant rate, independent increments.

Normal

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

R uses σ, even when the model is written $N(\mu, \sigma^2)$.

Lognormal

$$P(T \leq t) = \Phi\left(\frac{\ln t - \mu}{\sigma}\right).$$

Parameters describe $\ln T$, not T.

Exponential

$$f(t) = \lambda e^{-\lambda t}, \quad F(t) = 1 - e^{-\lambda t}, \quad R(t) = e^{-\lambda t}, \quad h(t) = \lambda.$$

$$P(T > s + t \mid T > s) = P(T > t).$$

Weibull

$$F(t) = 1 - e^{-\left(\frac{t}{\eta}\right)^{\beta}}, \qquad R(t) = e^{-\left(\frac{t}{\eta}\right)^{\beta}},$$

$$h(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}.$$

Shape	Meaning
$\beta < 1$	Decreasing hazard; early defects.
$\beta = 1$	Constant hazard; exponential.
$\beta > 1$	Increasing hazard; fatigue or wear-out.

$$F(\eta) = 1 - e^{-1} \approx 0.632, \qquad t = \eta(\ln 2)^{\frac{1}{\beta}}.$$

6. R distribution functions

Need	Prefix	Normal example
PMF/PDF height	d	dnorm(x, mean, sd)
Lower-tail probability	p	pnorm(x, mean, sd)
Quantile	q	qnorm(p, mean, sd)
Random sample	r	rnorm(n, mean, sd)

Replace norm with binom, pois, unif, lnorm, exp, weibull or gamma.

```
# P(X > x)
pnorm(x, mean = mu, sd = sigma, lower.tail = FALSE)

# P(a < X < b)
pnorm(b, mean = mu, sd = sigma) -
  pnorm(a, mean = mu, sd = sigma)

# c such that P(X > c) = alpha
qnorm(alpha, mean = mu, sd = sigma, lower.tail = FALSE)
```

Parameter reminders:

- pbinom(x, size = n, prob = p)
- ppois(x, lambda = rate * time)
- plnorm(t, meanlog = mu, sdlog = sigma)
- pexp(t, rate = lambda)
- pweibull(t, shape = beta, scale = eta)
- pgamma(t, shape = k, rate = lambda)

Use lower.tail = FALSE for reliability $P(T > t)$.

7. Estimation

For complete data,

$$\bar{x} = \frac{1}{n}\sum_i x_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i(x_i-\bar{x})}^{2}}.$$

Model	MLE
Normal	$\hat{\mu} = \bar{x}$, $\hat{\sigma} = \sqrt{\frac{\sum_{i(x_i-\bar{x})}^{2}}{n}}$
Lognormal	$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} \ln x_i$, $\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n(\ln x_i - \hat{\mu})^2}}$
Exponential, complete	$\hat{\lambda} = \frac{1}{\bar{x}}$

For censored exponential data, with r failures and total time on test $\mathcal{T} = \sum_i t_i$,

$$\hat{\lambda} = \frac{r}{\mathcal{T}}, \qquad \widehat{\mathrm{MTTF}} = \frac{\mathcal{T}}{r}.$$

Likelihood with status δ_i:

$$L(\theta) = \prod_i f(t_i;\theta)^{\delta_i} R(t_i;\theta)^{1-\delta_i}.$$

Complete-data plotting positions:

$$\hat{F}\big(t_{(i)}\big) = \frac{i}{n+1} \quad \text{or} \quad \frac{i-0.3}{n+0.4}.$$

Weibull plot:

$$\ln\big[-\ln\big(1-\hat{F}(t)\big)\big] = \beta \ln t - \beta \ln \eta.$$

Slope $= \beta$; intercept $a = -\beta \ln \eta$; therefore

$$\eta = e^{-\frac{a}{\beta}}.$$

Kaplan–Meier:

$$\hat{R}(t) = \prod_{t_j \le t} \frac{n_j - d_j}{n_j}.$$

Censoring causes no downward step, but it reduces the later risk set.

8. Sampling distribution and CLT

$$\bar{X} = \frac{1}{n}\sum_i X_i, \qquad E\big[\bar{X}\big] = \mu, \qquad \mathrm{SE}\big(\bar{X}\big) = \frac{\sigma}{\sqrt{n}}.$$

Situation	Standardisation
Normal population, σ known	$\frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$, exact
Normal population, σ unknown	$\frac{\bar{X}-\mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$, exact
Non-normal population, large n	$\bar{X} \dot{\sim} N\big(\mu, \frac{\sigma^2}{n}\big)$, CLT

Course rule of thumb: $n \geq 30$ is large, but strong skewness or far tails may need more.

Sample-size symbols

These formulas assume independent observations with population standard deviation σ known. They are exact for a normal population and approximate when using the CLT.

Symbol	Meaning
n	Required number of observations.
$\bar{X}$, μ	Sample mean and true population mean.
σ	Population standard deviation, not variance σ^2.
α	Allowed probability of falling outside the desired bound; e.g. 0.05.
$1-\alpha$	Desired probability of meeting the bound; e.g. 0.95.
d	Allowed upward deviation of the sample mean from μ (one-sided).
E	Allowed error in either direction from μ (two-sided).
z_p	Standard normal quantile: the value with probability p to its left, so $P(Z \leq z_p) = p$ for $Z \sim N(0,1)$. In R: qnorm(p).

Read the subscripts as a whole: $z_{1-\alpha}$ means the normal quantile at probability $1-\alpha$, not $z_1 - \alpha$. Similarly, $z_{1-\frac{\alpha}{2}}$ uses probability $1-(\frac{\alpha}{2})$. The σ after either quantile is a separate multiplication factor.

One-sided sample size: keep the probability of overshooting μ by at least d below or equal to α.

$$P(\bar{X} \geq \mu + d) \leq \alpha \quad \Rightarrow \quad n \geq \left(\frac{z_{1-\alpha} \times \sigma}{d}\right)^2.$$

Two-sided sample size: keep the sample mean within E of μ with probability at least $1-\alpha$. Split the allowed error probability equally between the two tails: $\frac{\alpha}{2}$ in each.

$$P(|\bar{X} - \mu| < E) \geq 1-\alpha \quad \Rightarrow \quad n \geq \left(\frac{z_{1-\frac{\alpha}{2}} \times \sigma}{E}\right)^2.$$

Desired probability $1-\alpha$	α	One-sided $z_{1-\alpha}$	Two-sided $z_{1-\frac{\alpha}{2}}$
0.90	0.10	$z_{0.90} \approx 1.282$	$z_{0.95} \approx 1.645$
0.95	0.05	$z_{0.95} \approx 1.645$	$z_{0.975} \approx 1.960$
0.99	0.01	$z_{0.99} \approx 2.326$	$z_{0.995} \approx 2.576$

For example, with $\sigma = 10$ hours and allowed error 2 hours at $1-\alpha = 0.95$:

- One-sided: $n \geq \left(1.645 \times \frac{10}{2}\right)^2 \approx 67.65$, so use $n = 68$.
- Two-sided: $n \geq \left(1.960 \times \frac{10}{2}\right)^2 \approx 96.04$, so use $n = 97$.

Square the entire fraction, then round n up to a whole number.

9. Final checks

- Did I define the random variable and its units?
- Is it a count or a continuous measurement?
- Does the distribution's support match the variable?
- Did I use variance or standard deviation correctly?

- Did I choose the requested lower, upper or two-sided tail?
- For a sample mean, did I use $\frac{\sigma}{\sqrt{n}}$?
- If σ is unknown, is t exact or only approximate here?
- Did I account for censored observations?
- Does the fitted hazard shape match the physical failure mechanism?
- Is the final number a probability, rate, time, variance or standard deviation, and are its units correct?